

Local Extrema on Surfaces

Local Maximum and Minimum

A function $f(x, y)$ has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) near (a, b) .

A **local minimum** at (a, b) means $f(x, y) \geq f(a, b)$ for all (x, y) near (a, b) .

Can a differentiable function f have a local maximum at (a, b) with $f_x(a, b) = 3$?

Theorem: Necessary Condition for Local Extrema

If f has a local extremum at (a, b) and the first-order partial derivatives exist there, then

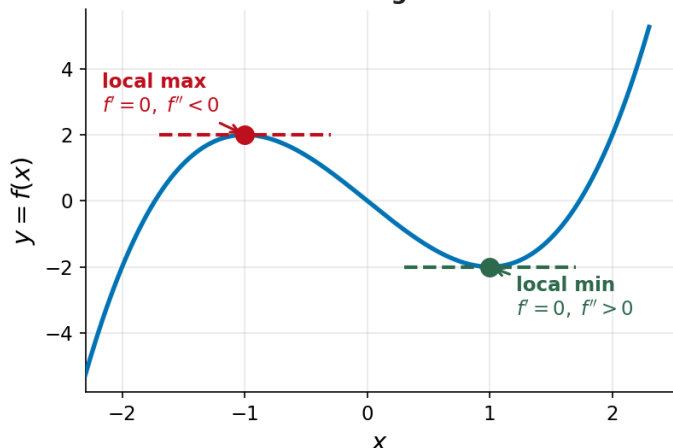
$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0$$

At a differentiable extremum, both partial derivatives must be zero — but zero partials alone aren't enough to guarantee an extremum.

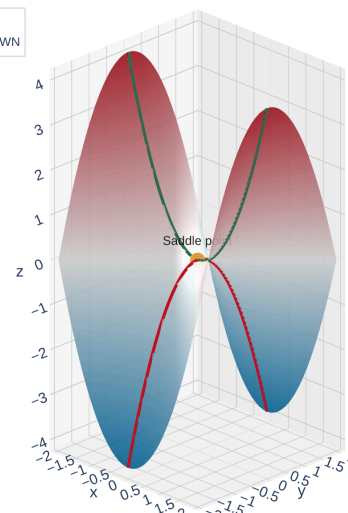
Optimization: From Calc I to Calc III

The big ideas carry over — but the **boundary** is now a curve instead of two endpoints, and **saddle points** are genuinely new.

Calculus I: Horizontal tangent at each extremum

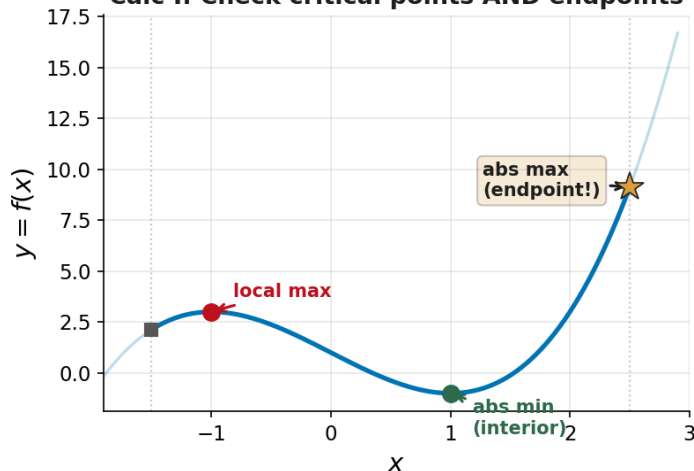


Along x: curves UP
Along y: curves DOWN

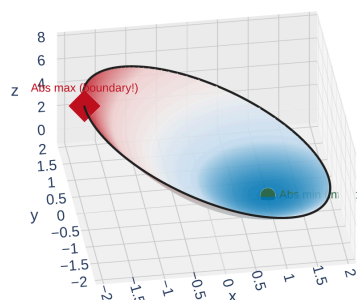


Interactive version

Calc I: Check critical points AND endpoints



Boundary curve
Abs min (interior)
Abs max (boundary)



Finding Critical Points

Critical Point

A point (a, b) is a **critical point** of f if:

- $f_x(a, b) = 0$ and $f_y(a, b) = 0$, **or**
- one (or both) partial derivatives do not exist there

Every local extremum occurs at a critical point — but not every critical point is an extremum!

Procedure for finding critical points:

1. Compute f_x and f_y
2. Set $f_x = 0$ and $f_y = 0$ simultaneously
3. Solve the system of equations for x and y

Example: Find the Critical Points

Example 1

Find the critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

Compute f_x and f_y , set both equal to zero, and solve the system.

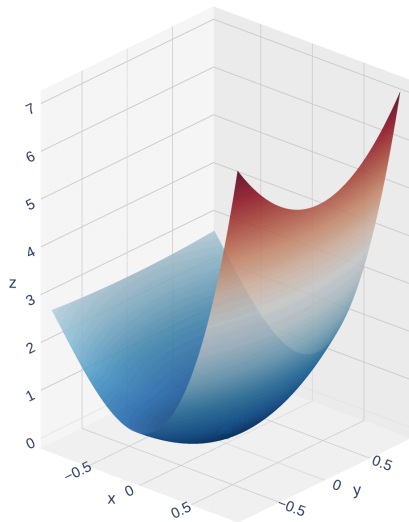
The Second Derivative Test

Second Derivative Test

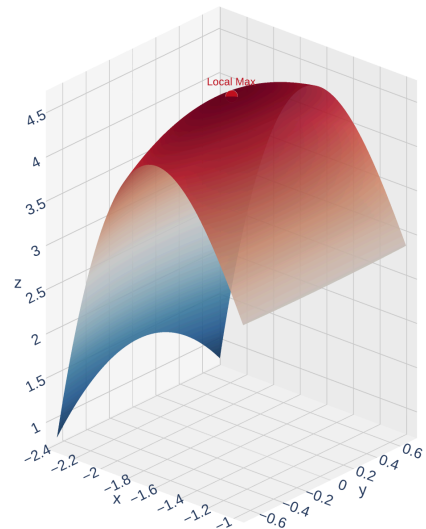
Suppose the second partial derivatives of f are continuous near (a, b) and $f_x(a, b) = f_y(a, b) = 0$. Define:

$$D = D(a, b) = f_{xx}(a, b) \cdot f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

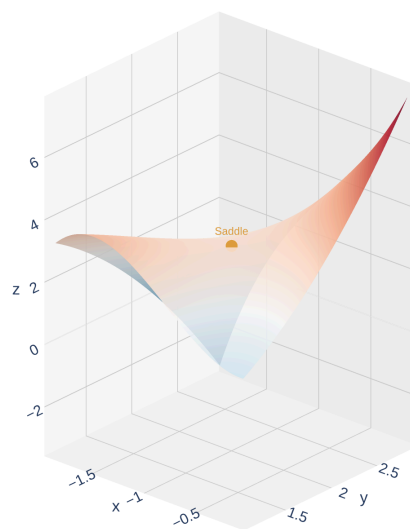
D measures **curvature agreement**: $D > 0$ means the surface curves the same way in all directions. $D < 0$ means f_{xy} dominates — the surface is **twisting**, curving up in one direction and down in another.



Interactive version



Interactive version



Interactive version

Quick Check: Use the Second Derivative Test

You've found a critical point and computed:

$$f_{xx} = -6, \quad f_{yy} = -4, \quad f_{xy} = 1$$

Compute D and classify. Local max, local min, or saddle?

Use $D = f_{xx}f_{yy} - (f_{xy})^2$. Check sign of D first, then f_{xx} .

Caution: When $D = 0$

Example 2

Show that $D = 0$ at the origin for $f(x, y) = x^4 + y^4$. Is $(0, 0)$ actually an extremum?

Compute all second partials at $(0, 0)$ and find D . Then argue from the formula for f .

Example: Classify the Critical Points

Example 3

Classify all critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

(We already found them: $(0, 0)$, $(-\frac{5}{3}, 0)$, $(-1, 2)$, $(-1, -2)$.)

Compute f_{xx} , f_{yy} , f_{xy} , then evaluate D at each critical point.

Your Turn: Find and Classify Critical Points

Example 4

Find and classify the critical points of $f(x, y) = x^3 - 3xy + y^3$.

Set $f_x = 0$ and $f_y = 0$. Express one variable in terms of the other and substitute.

The Extreme Value Theorem

Extreme Value Theorem (Two Variables)

If f is continuous on a **closed, bounded** set D in $\mathbb{R}^2$, then f attains an absolute maximum and an absolute minimum on D .

Closed set: includes all its boundary points (like $\leq$ vs. $<$)

Bounded set: fits inside some large disk (not stretching to infinity)

Cautionary notes:

- Absolute extrema may occur where f is **not differentiable**
- Absolute extrema may occur **on the boundary** (not just interior critical points)
- There can be infinitely many absolute extrema

Procedure for finding absolute extrema on a closed, bounded set D :

1. Find all **critical points** of f inside D
2. Find the extreme values of f **on the boundary** of D
3. The largest value from steps 1 and 2 is the absolute max; the smallest is the absolute min

Quick Check: Why Check the Boundary?

Where is the absolute max of $f(x) = x^2$ on $[-1, 2]$?

Example: Absolute Extrema on a Rectangle

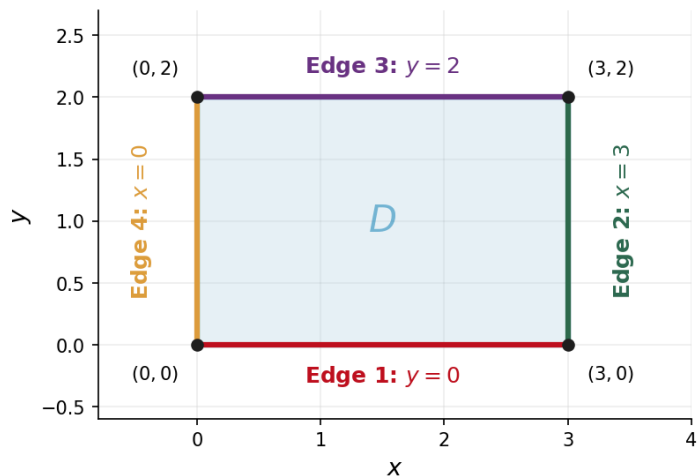
Example 5

Find the absolute maximum and minimum values of

$$f(x, y) = x^4 + y^4 - 4xy + 2$$

on the set $D = \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$.

Step 1: Interior critical points. Set $f_x = 0$ and $f_y = 0$:



$$f_x = 4x^3 - 4y = 0 \Rightarrow y = x^3$$

$$f_y = 4y^3 - 4x = 0 \Rightarrow x = y^3$$

Substitute $y = x^3$ into $x = y^3$ to find the interior critical points in D .

Checking the Boundary (Continued)

Step 2: Check f on each edge of the rectangle $[0, 3] \times [0, 2]$.

Evaluate f on all four edges. On each edge, one variable is fixed.

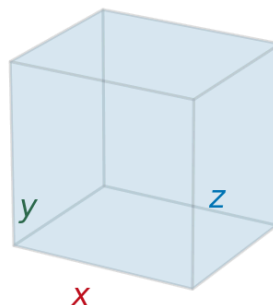
Application: Minimum Surface Area Box

Example 6

Find the dimensions of a rectangular box with volume 1000 cm^3 that has the smallest possible surface area.

Setup: Let the box have dimensions x, y, z (in cm).

- **Constraint:** $xyz = 1000 \Rightarrow z = \frac{1000}{xy}$
- **Minimize:** $S = 2xy + 2xz + 2yz$



Substitute $z = 1000/(xy)$ into S , then find critical points.

Homework

Section 14.7: 1, 3, 5-21 odd, 25, 33, 37, 41, 43, 45, 47, 49, 51, 55